

Exploring Travel Mode Choice in Victoria through Socioeconomic Predictors

Group W11G1

Noah Ryan
ryannr@student.unimelb.edu.au

Shi Jian Chan
shijian.chan@student.unimelb.edu.au

Gerard Czerwik
gerard.czerwik@student.unimelb.edu.au

Peiran Wang
peiranwang@student.unimelb.edu.au

Executive Summary

This study investigates the socioeconomic drivers of travel mode choice in Victoria using the 2023–2024 Victorian Integrated Survey of Travel and Activity (VISTA). Drawing on a sample of 2,404 individuals, the analysis examined 27 socioeconomic and demographic variables to determine which factors most strongly predict transport behaviour. There were two research questions (1) What socioeconomic factors are most important in predicting the Victorian population’s choice of travel mode? and (2) Are there distinct workforce profiles—by occupation, income, vehicle ownership, and suburb—that predict transport mode choice? A combination of correlation analysis, supervised machine learning, and clustering was applied. Vehicle ownership, household income, and home location emerged as the most influential predictors of travel mode, while age, education, and occupation were weakly associated. Machine learning models confirmed these drivers, with Random Forest achieving the strongest predictive performance. Clustering identified four distinct workforce profiles, highlighting sharp contrasts in reliance on private versus public transport across income and occupation groups. Findings reveal that car dependence dominates across all socioeconomic groups, with public transport concentrated among lower-income households. These results highlight the importance of targeted policy interventions, including expanding reliable public transport and reducing excessive vehicle ownership, to support equitable, efficient, and sustainable mobility in Victoria.

Introduction

Transportation is central to connecting people to employment, education and many more services worldwide (Tikoudis et al, 2024). Understanding the factors that drive people’s choice of travel mode is vital for designing efficient, equitable and sustainable systems. Research consistently highlights the role of socioeconomic factors, with international studies revealing household income, education, occupation and car ownership significantly influences travel behaviour (Halder & Mistri, 2025; Ko et al., 2019; OECD, 2024).

In Australia, the Bureau of Infrastructure, Transport and Regional Economics (BITRE) have identified challenges in transport access for different socioeconomic communities. Low-income households are three times more likely to depend on public transport, yet many live in outer suburban areas where services are infrequent and coverage is limited (BITRE, 2019). These patterns contribute to transport disadvantage, where inadequate access to reliable and affordable transport can constrain participation in work, study and community life (Currie et al., 2009).

Within Victoria, these concerns are amplified by rapid population growth. Melbourne is the fastest growing city in Australia, placing increasing demand on infrastructure and transport networks (Stevenson et al., 2024). Identifying the socioeconomic drivers of travel mode choice is critical for not only meeting mobility needs of the expanding population but also guiding sustainable infrastructure investment and supporting Victoria’s long-term climate targets (DTP, 2023).

Two interrelated research questions have been developed to investigate these patterns:

1. What socioeconomic factors are most important in predicting the Victorian population's choice of travel mode?
2. Are there distinct workforce socio-economic profiles — by occupation, income, vehicle ownership, and suburb — that predict transport mode choice?

This study draws on recent personal and household data from the Victorian Integrated Survey of Travel and Activity (VISTA), surveyed from a large, randomly selected sample of 2,404 Victorians during 2023-2024 (Transport Victoria, 2025). The combined dataset used includes 27 variables capturing both travel behaviour and sociodemographic characteristics. Key socioeconomic variables of interest include age, occupation, education level, household income, home suburb, and vehicle ownership. By identifying the strongest socioeconomic predictors of travel mode, infrastructure planning and investment can be more targeted to where they will have the greatest social and environmental impact.

Methodology

Data Processing

The preprocessing stage consolidated and cleaned multiple VISTA 2023–2024 datasets to create a single dataset, suitable for correlation and machine-learning analyses. Household-, person-, and trip-level data (including journey-to-work and education records) were merged through unique household and person identifiers, allowing socio-economic characteristics to be linked directly with individual travel-mode choices.

Data cleaning involved removing incomplete records, standardising variable types, and assigning “Unknown” to missing income, car ownership and travel mode values to maintain representativeness. Categorical variables with excessive granularity were collapsed into broader, interpretable groups. Household income was classified as Low (< \$800/week), Medium (\$800–1,999/week), High ($\geq$ \$2,000/week), or Unknown. Vehicle ownership was reduced to 0, 1, or 2+ vehicles; age into five bands (0–17, 18–24, 25–44, 45–64, 65+); and travel modes into Car Driver, Car Passenger, Public Transport, Active Mode, and Other. Regional and dwelling variables were simplified to Metropolitan vs Regional and House vs Apartment/Other.

Only features relevant to the research questions—income, vehicle ownership, employment, education, age, gender, region, household size, trip purpose, and travel mode—were retained, while identifiers and redundant fields were dropped.

These preprocessing steps ensured a consistent and interpretable dataset aligned with the project's objective of predicting travel-mode choice from socio-economic factors. The categorical collapsing reduced data sparsity and simplified complex survey responses, improving model performance and interpretability. Handling missing values through explicit labelling rather than deletion preserved data diversity and prevented urban or low-income bias. Simplified groupings (e.g., income tiers, vehicle count, age bands) also support clearer correlation patterns and stronger explanatory power for supervised learning. Overall, the chosen preprocessing techniques balanced statistical robustness and real-world interpretability, providing a reliable foundation for subsequent analysis and modelling of Victorian travel behaviour.

Correlational Analysis

Socioeconomic and demographic features that were relevant for analysis included Age stage, Occupation, Education level, Household income, Home suburb, and Vehicle ownership. These variables were compared with the broadly categorised primary travel modes and specific primary travel modes to determine whether detail differences are present during correlation analysis.

Research Question 1 (RQ1) was explored by examining the association between these socioeconomic factors and travel mode correlation analysis was conducted using Chi-square test of independence, Cramér's V, and Normalised Mutual Information (NMI). This approach was chosen because the dataset comprised of categorical and binned continuous variables, restricting ability to use Pearson's r. The chi-square test of independence identifies whether each predictor was statistically associated with travel mode. Cramér's V measures the strength of the associations between predictors and travel mode (scaled 0-1). NMI assessed how much knowing a socioeconomic variable reduced uncertainty about travel mode. Because raw mutual information values are influenced by the complexity of each variable, NMI was used to scale results between 0-1. The 'min' normalisation method was applied, by dividing mutual information by the smaller entropy of the two variables. This conservative approach avoids overestimating association strength when one predictor has fewer categories than travel mode, allowing for fair comparisons.

Machine Learning

To evaluate the predictive performance and identify important features, supervised models were developed. The algorithms tested include Logistic Regression, Random Forest and Decision Trees to develop a predictive model that could correctly classify members of the dataset into either a general travel mode group (Private, Public, Active, Other) or specific travel modes (Vehicle Driver, Bus, Tram, etc.). The algorithms were specifically chosen due to their computational efficiency, easy interpretability and powerful predictive capabilities allowing key travel patterns to be accurately and efficiently modelled.

Other machine learning models like XGBoost and Neural networks were considered due to their ability to reliably predict categorical variables, however, Neural networks require much larger datasets to reduce overfitting and can be very computationally expensive (Donges, 2023). Similarly, XGBoost requires extensive tuning to perform at a similar level as Random Forest (XGBoosting, 2024) which was not viewed as an efficient use of time. Most importantly, both Neural Networks and XGBoost are complicated algorithms which are difficult to interpret and intuitively enhance. This limits their usefulness in analysing human data, where meaningful context and transparency are critical to understanding a model's predictions.

To address the RQ1, we focused on exploring the broad travel categories from the sample and incorporated the most amount of data points. Exploratory checks found significant bias in data distribution of travel modes (shown in Figure 1). Decision Trees and Random Forest were tested as they handle bias effectively, especially with balanced class-weight option, and deriving complex relationships between a multitude of potential predictors and travel mode. These models also remain highly understandable, allowing models to be decomposed and tuned by intuitive insights and human input. Random forest, an ensemble method build from many decision trees, extends these advantages by reducing the risk of overfitting and improving model robustness (IBM, 2025).

Given the data characteristics, the parameters throughout each model applied a fixed random state to ensure reproducibility. The maximum tree depth was limited to 15 to prevent overfitting. For the Random Forest model, 100 estimators were used to capture sufficient information through repeated calculations without overburdening the process with computation.

Feature selection was carried out to identify the most relevant socioeconomic predictors whilst avoiding redundancy between closely related variables. Some features captured similar information at different levels of specificity e.g. home suburbs and general region categories. To prevent models from being biased by overlapping information, complementary features were compared and assessed through comprehensive analysis. Gini index proportions were used to evaluate important predictors to support RQ1.

Following feature analysis, hyperparameter tuning was conducted on the top-performing models to optimise predictive performance. K-fold cross-validation was used to systematically evaluate parameter variations within the machine learning model, ensuring results were not dependent on a single train-test split. Model

accuracy was prioritised as the primary evaluation criterion, with the average accuracy across folds providing a robust measure of generalisability. This approach allowed the selection of parameter settings that balanced predictive strength with stability, reducing the risk of overfitting while maximising model reliability.

Clustering

This stage of analysis addresses the third research question: *Are there distinct workforce socioeconomic profiles-by occupation, income, vehicle ownership and suburb – that predict transport mode choice?* By only focusing on workforce, it allows for more detailed investigation to take place. This separation was necessary because students and workers face different travel behaviours and may obscure important patterns (Papafoti et al., 2025). Only employed individuals were included in workforce sample.

Clustering was based on four predictors: occupation (five categories), household income (three groups), vehicle ownership (three groups) and home suburb. These variables were selected based on previous findings from correlation and machine learning analysis. Since all predictions are categorical, categories were transformed into numerical representations before visual analysis. Standardisation was performed using Z-score scaling to prevent single features from dominating the clustering outcome.

K-means clustering was selected due to its scalability and interpretability, dividing workers into subgroups based on how similar they were socioeconomically. Alternative approaches like hierarchical clustering were less suited due to the dataset size. The optimal number of clusters (k) was determined using the elbow method and confirmed with silhouette analysis. This ensured the clusters identified are both statistically robust and substantially representative of different socioeconomic groupings relevant to travel behaviour. Principal Component Analysis (PCA) was applied to reduce noise in the data and produce a visualisation of the clusters.

Once clusters were identified, they were profiled against the broad transport mode categories to highlight dominant patterns of transport behaviour within each socioeconomic group. Broad transport mode categories were used because unlike correlation analysis, where detailed travel mode produces more meaningful associations, broad categories prevent unnecessary splitting of clusters and reduces overlapping subgroups which improve interpretability. To interpret each cluster, cross tabulations were performed to identify key socioeconomic variables that shaped travel mode distributions.

Results Exploration and Analysis

Preliminary Analysis

As shown in Figure 1, the distribution of broad travel modes across the whole sample reveals a strong bias towards private transport. When split into student and workforce sub-samples in Figure 2, private transport remains the majority mode. Public transport use is much higher in student sample than workers.

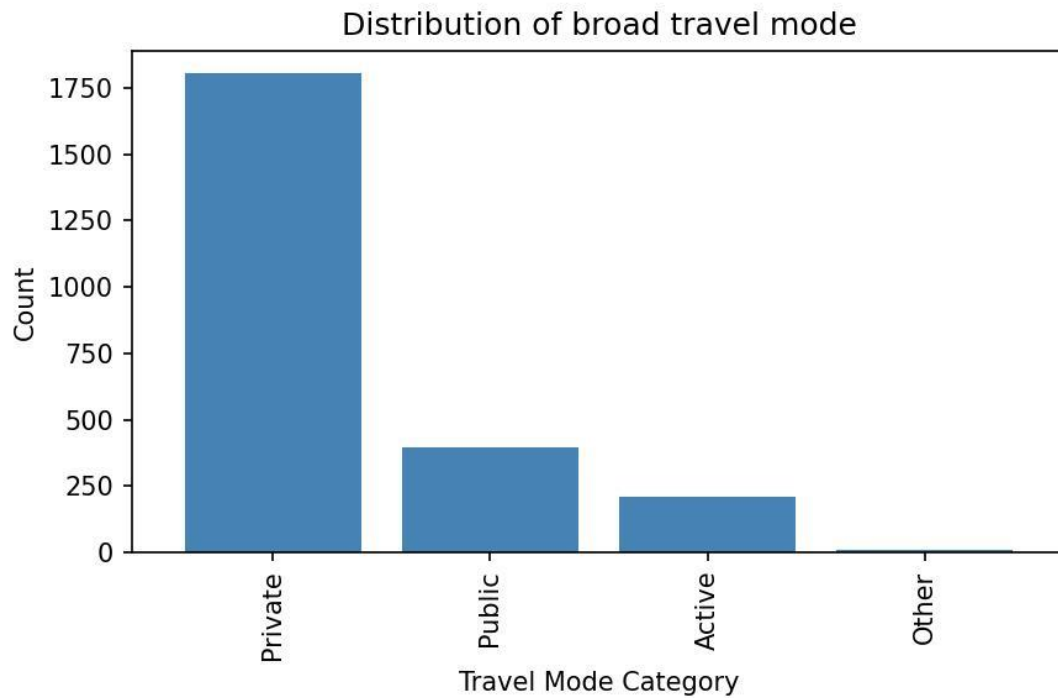


Figure 1: Sample distribution of broad primary travel mode.

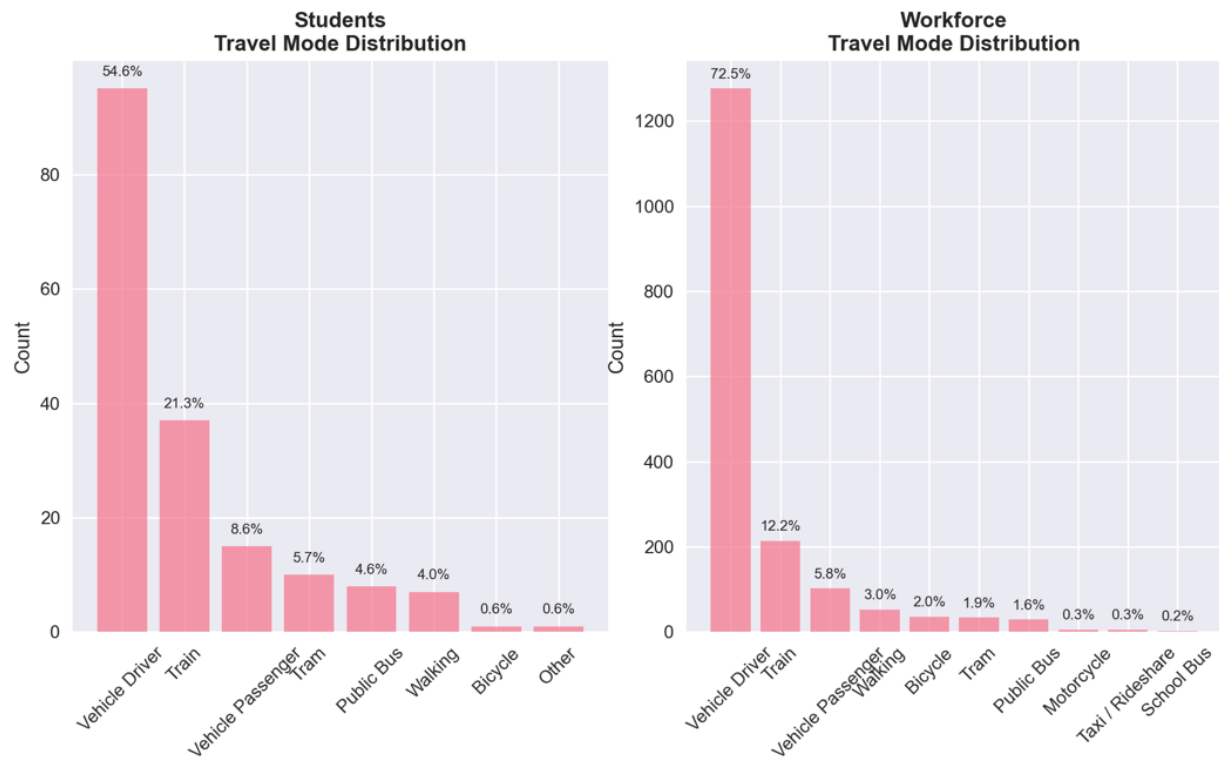


Figure 2: Distribution split between Student and Workforce samples.

Correlational Analysis

When comparing socioeconomic variables with the detailed categories of primary travel mode, chi-square test showed significant associations for all observations. The association between two variables were calculated using Cramér's V, with the Normalised Mutual Information for each variable displayed in Table 1. Vehicle ownership, home location and household income were moderately associated with travel mode choice, while education, occupation and age stage showed weak associations.

Table 1: Cramér's V and Normalised Mutual Information computed between predictor and travel mode.

Predictor	Cramér's V	NMI
Vehicle Ownership	0.29	0.21
Home Location	0.26	0.20
Household Income	0.24	0.18
Education Status	0.17	0.12
Occupation	0.15	0.11
Age Stage	0.13	0.10

Machine Learning

Figure 3 displays a heatmap of feature importances based on Gini index. The results indicate that several socioeconomic variables contribute to predicting broad travel mode choice. Trip duration showed high importance across all models. Home location (homelga) and household income (inc_group) emerged as two of the strongest predictors across models. Personal income, household income category, subregion, total vehicles owned and sex all contributed as minor factors to each model.

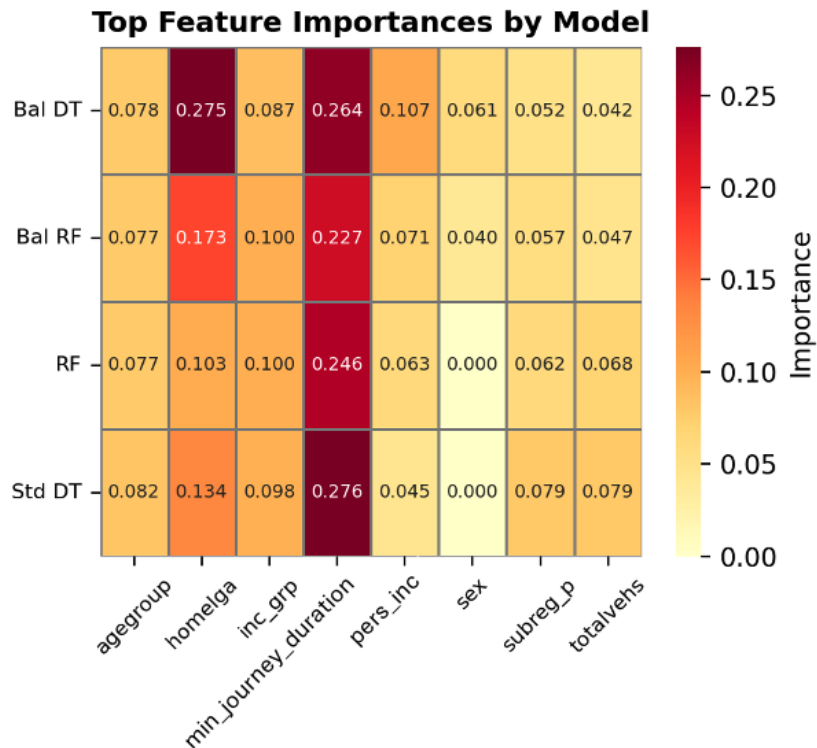


Figure 3: Heatmap comparisons of feature importance in each model using Gini Index

When considering the Victorian population as a whole and performing feature analysis to clarify the most important contributing factors, Random Forest provided the best overall performance in terms of its accuracy, precision, recall and support tested across all factors and compared using independent pairwise comparisons of processed and non-processed data types. Pairwise comparisons were completed on the household income groups, vehicle ownership categories, suburb and region classifications and comparing student indicators with age features.

Table 2: Table depicting one of the comparisons performed between the processed and raw vehicle ownership features where Random Forest performed the best of all the models.

Classification Report - Random Forest | Optimum Feature: total vehicles (int)

Target Var (Broad Travel Mode)	Precision	Recall	F1-Score	Support
Active	0.58	0.27	0.37	41.0
Other	0.00	0.00	0.00	1.0
Private	0.80	0.93	0.86	361.0
Public	0.44	0.23	0.30	78.0
Macro Avg	0.45	0.36	0.38	481
Weighted Avg	0.72	0.76	0.73	481

When performing hyperparameter tuning via K-fold cross validation to maximise the average accuracy of the random forest model, a max depth of 20, with 100 estimators and a balanced subsample class weight approach can push the accuracy to 0.7963 with a cross-validation score of 0.8164.

Clustering

The elbow plot displayed in Figure 4 shows a steep decline in within-cluster variation (inertia) up to $k=4$, where improvements after are less extreme. Selecting $k=4$ provided an optimal balance, capturing sufficient clustering structure whilst maintaining interpretability and simplicity.

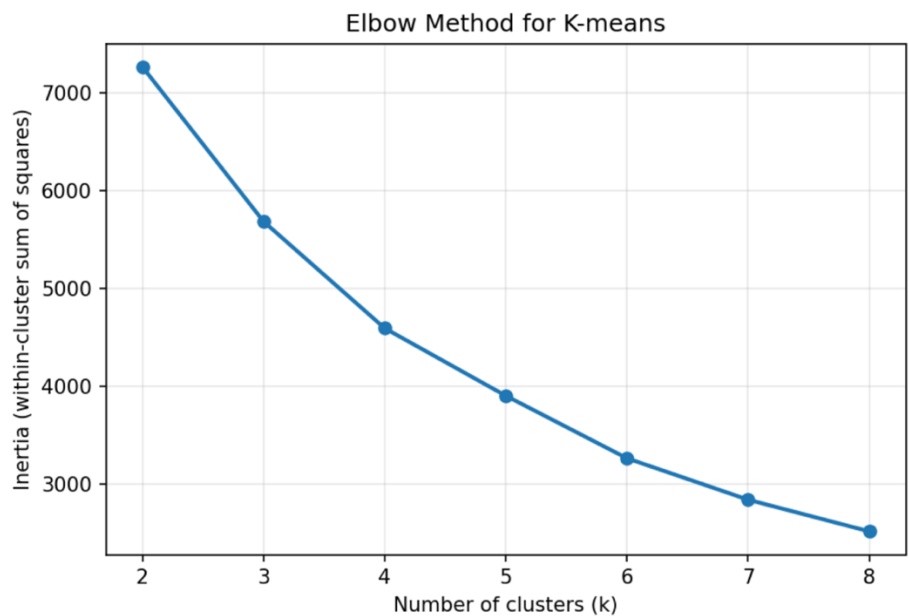


Figure 4: Elbow method used to determine optimal number of clusters for workforce.

The distribution of travel modes was calculated across the 4 workforce clusters and displayed in Table 3. Private transport mode remains the most dominant mode in all clusters. The greatest variation between clusters is seen in Public transport mode. Cluster 0 exhibited the highest reliance on public transport use and the lowest Private transport use among clusters. In contrast, Cluster 1 and 3 had highest reliance on Private transport, and lowest on Public. Cluster 2 revealed an intermediate profile with a relatively moderate prevalence of Public Transport.

Table 3: Distribution of broad travel modes across workforce clusters.

Cluster	Active (%)	Private (%)	Public (%)
0	8	63	29
1	4	89	7
2	6	77	17
3	2	88	10

Figure 5 visualises the four clusters using Principal Component Analysis. PCA1 and PCA2 was able to account for 58% of the total variance, which provides a sufficient basis for plotting clusters in two dimensions while preserving most of underlying structure. Although there are some visible overlaps between clusters, the groupings each cluster forms recognisable groupings, reflecting differences in transport behaviour.

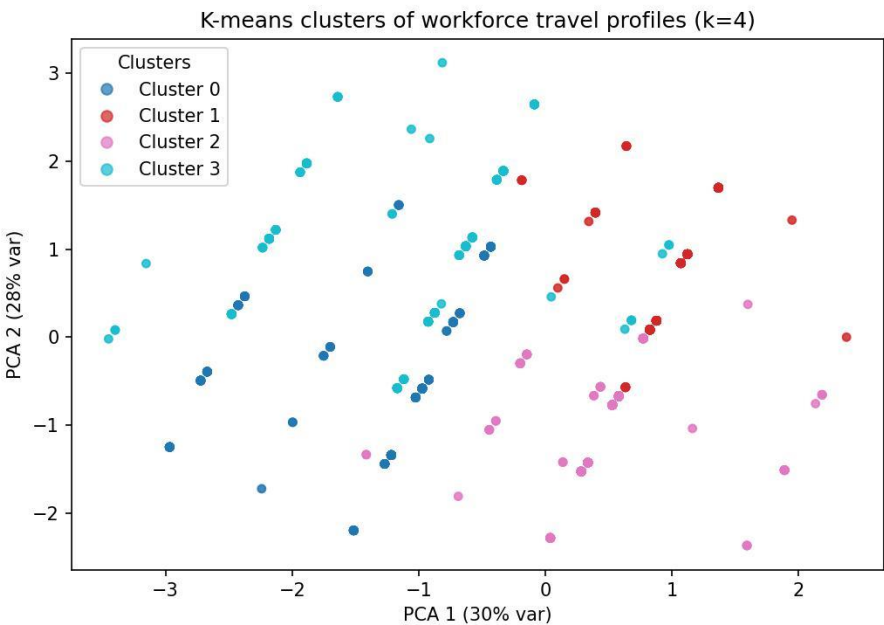


Figure 5: Cluster Visualisation using Principal Component Analysis (PCA)

Manual analysis on key industries in each cluster revealed significant differences. Cluster 0 primarily comprised of service, and lower income workers that had lower prevalence of vehicle ownership, which is reflected by the high dependence on public transport. Cluster 1 reflected professional and management roles who have higher income and more than 2 accessible private vehicles. Cluster 2 comprised of a mix of technical and semi-professional roles with moderate income and vehicle ownership. Cluster 3 was dominated by trade and outer suburban workers with high car ownership, often having access to more than 2 vehicles.

Discussion and Interpretation

Upon analysis, several interesting and insightful patterns were revealed across correlation analysis, machine learning models, and clustering that shed light on how socioeconomic factors shape travel behaviour in Victoria.

- Dominance of private transport

Across the sample private transport was the most common mode of travel, even persisting across student and workforce subgroups, and workforce clusters. The heavy reliance on private vehicles was interesting to observe, especially given the scale of its impact. Even in public transport accessible areas, car use remained the travel mode for majority of individuals. This can raise interesting points for reasons why individuals choose private transport over public. Chwasta (2024) also reported that Victoria's rate of vehicle ownership has increased more than population growth in recent years.

From the analysis the strongest consistent predictors were vehicle ownership, home location and household income. Clustering played a critical role in highlighting the dynamics of vehicle ownership, showing that household with multiple cars and lower income individuals all exhibited high, majority reliance on private transport. However, with higher income were able to access multiple vehicles. Home location is an important factor, although individuals in inner suburbs have dense public transport networks, there is only a slight increase in reliance in public transport. Outer suburb residence has highest dependence which can be contributed to the distance and lack of service coverage between their place of work, which revealed through clustering commonly consists of trade workers.

This highlights how important of car ownership is to individuals in Victoria, further raising potential issues with current infrastructure and public transport systems that discourage individuals to use public transport as a primary travel mode. Furthermore, this echoes long-standing concerns about Melbourne's urban sprawl and distribution of services. The transport disadvantage identified by Currie et al. (2009) remains a critical barrier, as even lower-income households are still reliant on cars for transport.

- Machine learning insights

Supervised machine learning confirmed these structural drivers but also revealed challenges in accuracy. Across the tested models, Random Forest consistently outperformed other approaches in terms of precision, recall, and F1-score. Additionally hyperparameter tuning increased Random Forest accuracy and validation score which demonstrates that the model learned meaningful patterns in the data,

A consistent constraint in the models was the uneven prediction strength. Most predictions can accurately predict private transport, whilst public and active transport were significantly less reliably classified. This imbalance mirrors the dataset itself, when private travel biases the sample, algorithms would naturally learn to predict it more confidently. This finding is valuable because it demonstrates that predictive modelling alone is insufficient to capture less frequently occurring behaviours like public transport or active travel. Even though algorithms initially chosen are best suited to overcome biased data, the imbalance of car dependence is still strong.

- Clustering in workforce

The PCA visualisation showed relatively clear but overlapping clusters, which is an important finding. While the k-means algorithm forces individuals into discrete groups, the overlap in two-dimensional space reflects that real-world travel behaviour is more fluid than discrete grouping may suggest. Variables that were not measured in the dataset may have been useful to gain a comprehensive view such as workplace location. This insight is valuable for policy and planning, as it demonstrates that interventions cannot treat groups as entirely homogenous. Even within a cluster, variation is still present.

· Implications

These findings highlight the need for flexible, targeted interventions and investment of resources to improve infrastructure planning and sustainability goals. Instead of focusing on the increased volume of private vehicles, it is important to consider why Victorians remain dependant on this form of transport.

Population growth is a key contributor, placing pressure on infrastructure demands and service coverage. For lower-income and low vehicle ownership groups, who are more reliant on public transport, addressing unmet needs is critical. Condren (2024) notes that public transport usage has fallen 42% since 2018, largely due to network inefficiencies such as infrequent services, with many buses running only every 30-60 minutes in suburbs despite high demand. At the same time, higher-income and car dependent groups present a different challenge, reducing transport emissions. Road transport remains the largest source of emissions in Victoria (Department of Transport and Planning, 2023). Policies that can increase incentives to purchase low-emission vehicles alongside strategies to encourage fewer cars per household can play a role in meeting Victoria's Net Zero by 2045 objectives.

Limitations and Improvement Opportunities

While some meaningful patterns emerged from the analysis, several limitations constrain the reliability and generalisability of the results. Although the VISTA dataset is relatively large, it contains significant imbalances in variables and certain categories. For example, the student sample ($n=174$) is much smaller than the workforce sample ($n=1759$), limiting representation in the analysis. Similarly, the simplification of travel modes into single categories overlooks the complexity of multi-modal trips, where some individuals may combine car, train and walking within the same journey. Vehicle ownership, although one of the strongest predictors, also lacks detail regarding affordability or vehicle type (second-hand, new, private or company car), which could significantly influence ownership decisions.

Methodological constraints can also influence outcomes. Most variables were treated as categorical, including income which was reduced into Low, Medium, High groups. While this improved interpretability, it sacrificed details that may reveal subtle but meaningful socioeconomic patterns. In machine learning, feature selection emphasised pairwise comparisons of predictors, particularly those representing similar information (eg. Suburb vs Broad Regions). However, a more systematic multivariate feature evaluation was not performed, which may explain why some models plateaued at ~72% F1 score, indicating that some features were not fully captured.

Clustering introduced additional constraints in its assumptions. K-means clustering assumes spherical, evenly sized clusters, which can oversimplify complex behaviours and blur fluid real-world patterns. Furthermore, some occupational groups such as agricultural or service workers were underrepresented in the dataset, further limiting ability to generalise findings across the entire Victorian workforce.

Incorporating continuous scales such as income, and more detailed details like vehicle information and work location can enhance the strength of patterns observed in the analysis. Tackling these limitations can help enhance predictive accuracy and allow for stronger predictions to be made.

Conclusion

This study investigated the socioeconomic drivers of travel mode choice in Victoria using the 2023–2024 VISTA dataset. Results show that some variables play a much stronger role in shaping mobility than others. Addressing Research Question 1, household income, vehicle ownership, and home location consistently emerged as the strongest predictors of travel mode choice. These reflect household resources and geography rather than individual attributes like age or education, which had weak associations. For Research Question 2, clustering identified four distinct workforce profiles. Lower-income service workers with limited car access relied most on public transport, while high-income professionals and trades workers were highly car-dependent. Technical and semi-professional roles showed a mixed profile of car and public transport

use. This indicates that occupation interacts with income, car ownership, and location to shape workforce travel behaviours. Together, these findings highlight key policy implications. Tackling car dominance requires both improving the quality and affordability of alternatives and reducing excessive car ownership in affluent households. Expanding reliable, frequent, and accessible public transport is essential for those most dependent on it. Finally, sustainability strategies must reduce emissions and congestion to align transport planning with Victoria's net-zero targets. In sum, resources and geography remain central to transport behaviour in Victoria. Recognising these structural drivers provides critical evidence for building more equitable, efficient, and sustainable mobility systems in a rapidly growing city.

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